



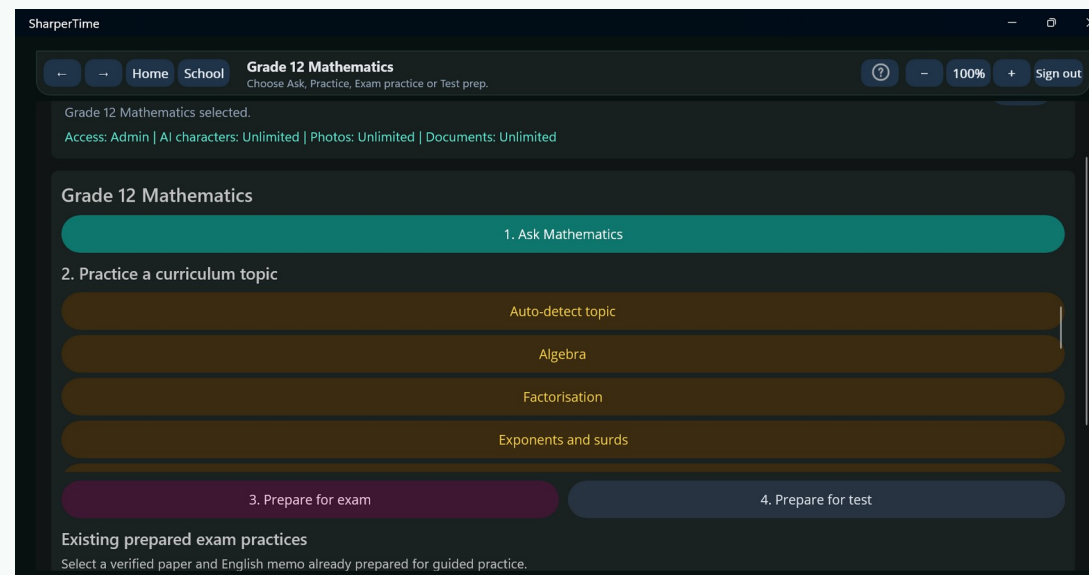
SharperTime

AI-powered Grade 12 Mathematics tutoring for African learners

A working Windows and Android platform for practice, hints, diagnostic marking, multimodal input and lesson videos.

Sharper X (Pty) Ltd
Founder: Lesego Mohaule
Johannesburg, South Africa

Application deck for Google Africa Applied AI Lab



The problem

Affordable, personalised mathematics help is still difficult to access.

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Many learners need help exactly when they are studying alone.

- Private tutoring is effective but too expensive for many families.
- School mathematics is multimodal: equations, working, graphs and diagrams.
- Generic AI tools do not provide a structured learning flow, marking and feedback.
- Learners need immediate correction, not only final answers.

R35–R50

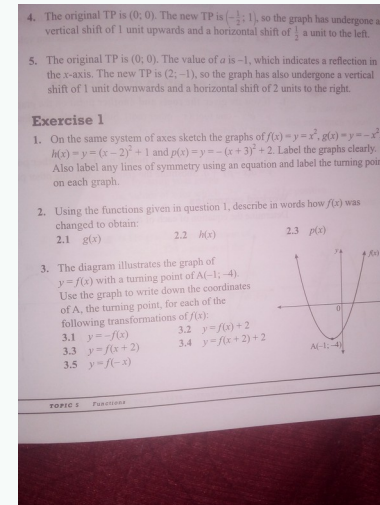
Target 30-day price

Grade 12

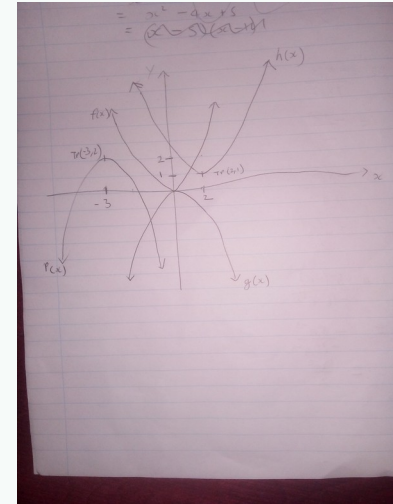
Initial launch focus

**Local →
Africa**

Expansion path



Textbook question



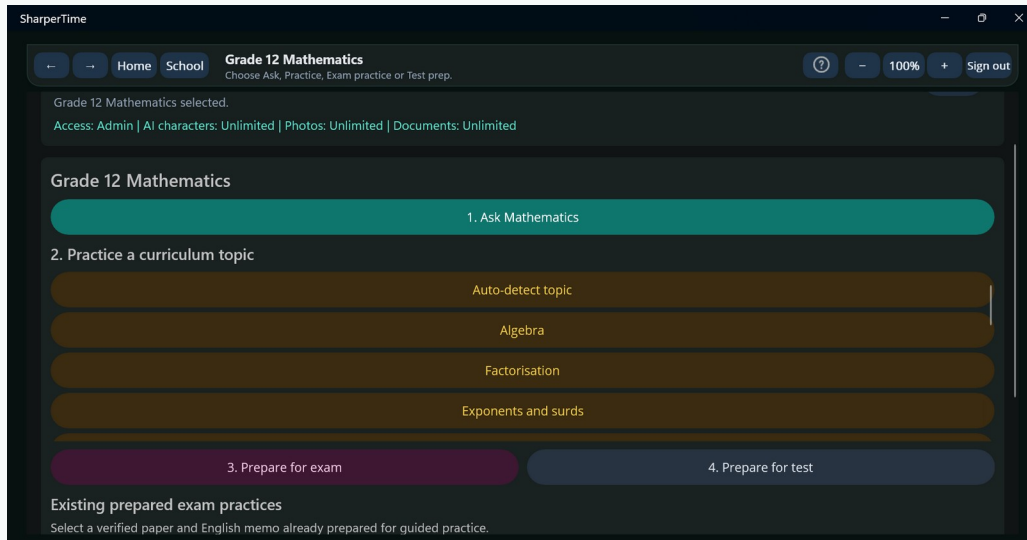
Handwritten graph

The solution

A focused AI tutor built around real mathematics workflows.

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SharperTime combines structured learning with multimodal AI.

- **Ask** Learners ask mathematics questions naturally.
- **Practice** Difficulty-based practice with hints and guided feedback.
- **Mark** The app checks answers and explains mistakes.
- **Capture** Textbook images, handwritten work, voice and graphs.
- **Learn** Short lesson videos connected to revision and practice.

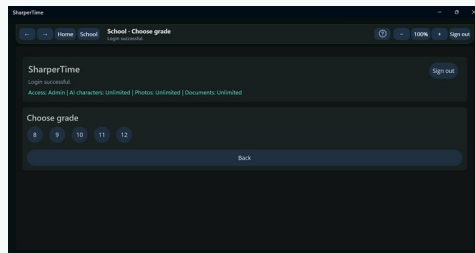
Not a generic chatbot — a learning platform designed for a focused Grade 12 Mathematics launch.

Product already working

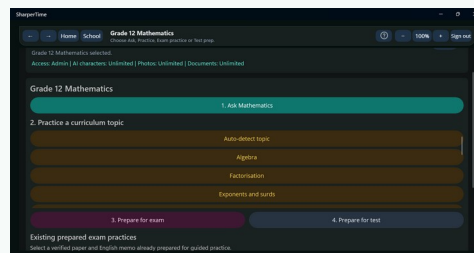
A real app experience has been recorded and demonstrated.

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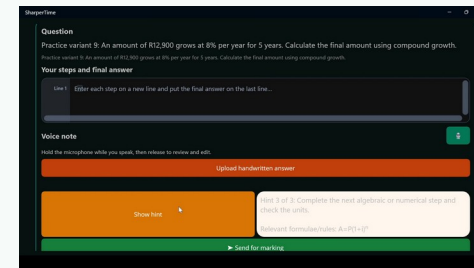
Google Africa Applied AI Lab



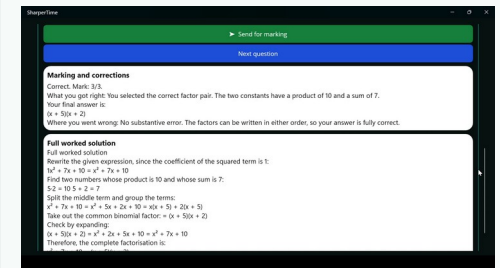
Choose grade



Select topic / mode



Get a hint



Receive marking

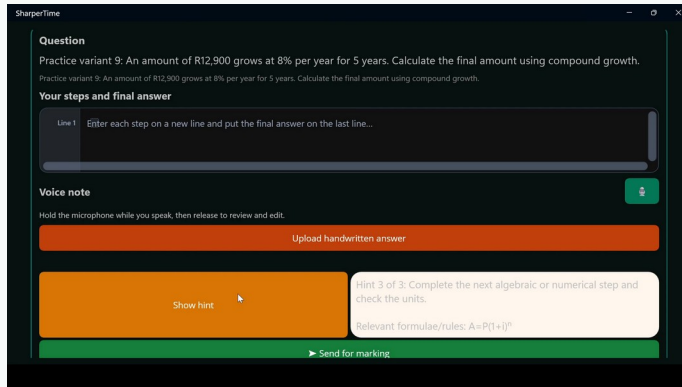
The product demo shows the actual app: topic choice, difficulty selection, hints, answer entry and diagnostic correction feedback.

AI tutoring and practice workflow

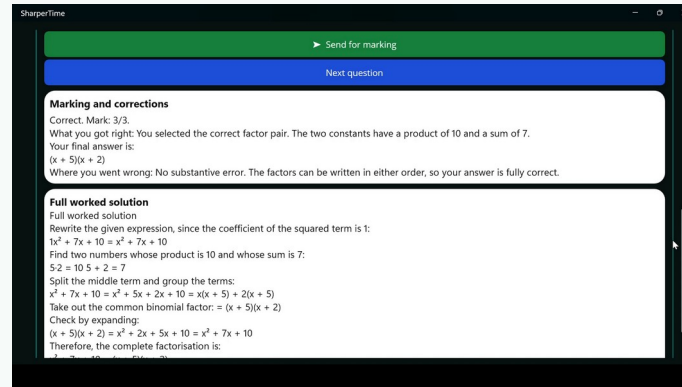
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The learner does the work; the app guides, checks and explains.

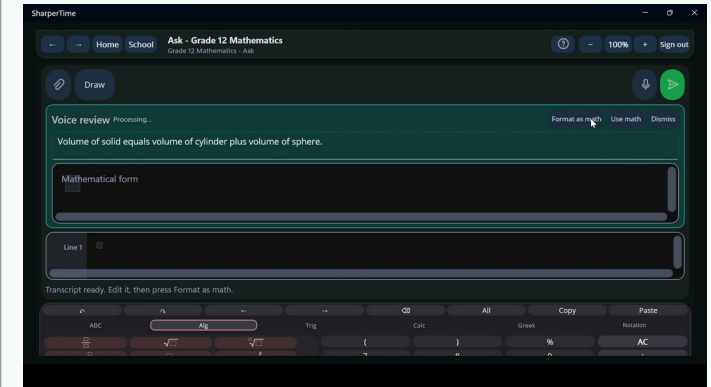
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Optional hints and formulas



Diagnostic marking



Review before sending

1

Select topic and difficulty

2

Attempt a real question

3

Ask for hint when needed

4

Receive marking and correction

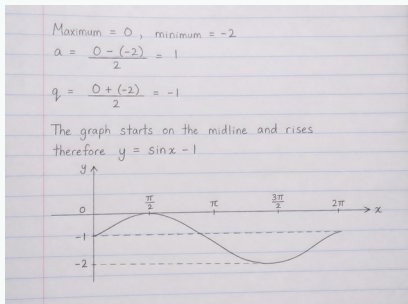
The current product focuses on user-selected difficulty, hints and corrective feedback rather than giving full solutions as the default learner experience.

Multimodal AI learning

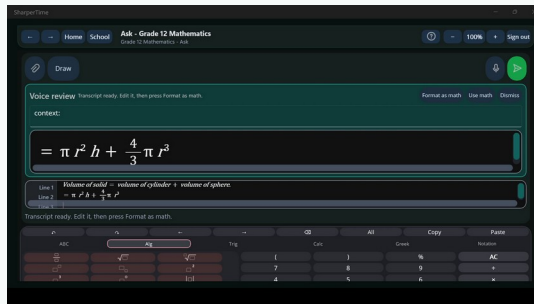
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SharperTime works with the way learners actually produce mathematics.

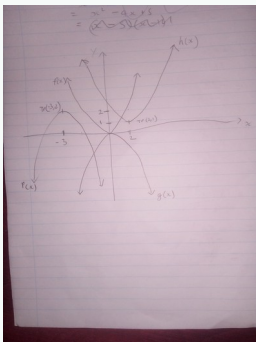
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Handwritten solution



Voice to maths



Hand-drawn graph

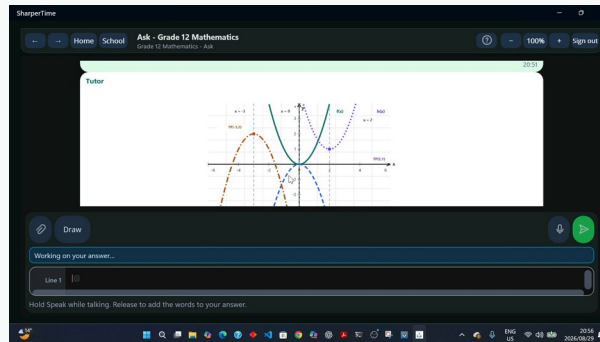


Image upload and analysis

Inputs

- Typed mathematics
- Voice converted into mathematical notation
- Textbook and photo uploads

- Handwritten working and graphs

Outputs

- Editable transcription and confirmation
- Mathematics reasoning and marking
- Specific feedback and correction
- Graph-aware responses

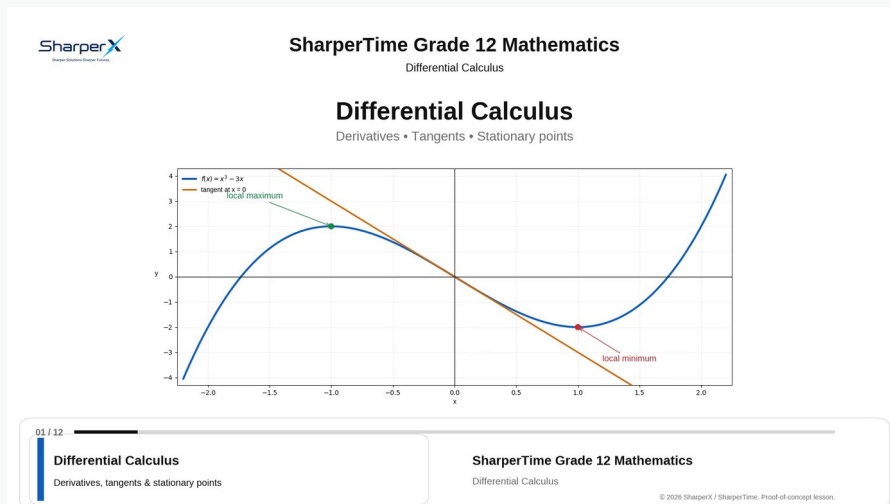
This matters because school mathematics is not just a text-chat problem.

Lesson-video content engine

Google AI tools can help create, refine and expand short CAPS-aligned lessons.

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Differential Calculus lesson sample

SharperTime Grade 12 Mathematics
Differential Calculus

Now write the tangent equation

$$y - y_1 = m(x - x_1)$$

Point: (0, 0) and gradient: $m = -3$

$$y - 0 = -3(x - 0)$$
$$y = -3x$$

The derivative supplies the gradient; the graph supplies the point.

08 / 12

Equation of the tangent
Use the point and the derivative gradient

SharperTime Grade 12 Mathematics
Differential Calculus

© 2020 SharperX / SharperTime. Proof-of-concept lesson.

Worked-example teaching style

Initial focus

High-school Mathematics,
starting with Grade 12.

Content expansion

All topics, subtopics and sections
for each grade.

Future scope

Other high-school subjects and
TVET engineering.

Why Google AI can accelerate SharperTime

SharperTime

Google AI can become materially impactful across product quality and scale.

Google Africa Applied AI Lab

Gemini

- Stronger multimodal maths understanding
- Better handling of handwriting, images, diagrams and graphs
- Richer explanations and tutoring interactions

Gemma / efficient models

- Lower-cost routing for routine tasks
- Potential support for low-connectivity environments
- Reusable feedback and question-classification workflows

Google tooling

- Lesson-video creation and content iteration
- Question refinement and evaluation workflows
- Support for Apple/Mac expansion and cloud-scale delivery

Goal: improve multimodal accuracy, question quality, lesson-content scale, latency and cost while keeping the learner experience simple.

Checkout offer at launch

One clear checkout option combines tutor access, lesson-video access and a defined AI usage allowance.

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R35 plan

30 days of combined Grade 12 Mathematics tutor access and lesson-video access, together with a standard AI usage allowance with a service value of R5.

The R35 plan includes tutor access and lesson videos; the included standard AI usage allowance has a service value of R5.

Learners may top up the AI usage balance on a pay-as-you-go basis during the month. Top-ups increase the in-platform AI usage balance and do not create a separate access product.

Go-to-market

Start locally, prove renewals, then expand outward.

Beachhead

- Marikana and surrounding communities in North West
- Gauteng school and community outreach
- Direct school visits, demos and local trust-building

Digital channels

- YouTube ads and lesson videos
- X / Twitter and Facebook
- Website launch page, product demo and community sharing

Launch path

- September launch on website and Google Play Store
- Focused on Grade 12 Mathematics first
- Expand across Africa after the core model is proven

Product roadmap

SharperTime

Focused Grade 12 Mathematics launch first, then disciplined expansion.

Google Africa Applied AI Lab

Now

Windows and
Android app
Grade 12
Mathematics

Sep 2026

Launch on website
and Google Play
Store

2027

Add other grades
and high-school
subjects

Next

Expand to Apple,
Mac
and more devices

Platform

Add SIP live
tutoring
and a shared
canvas

Later

TVET engineering
and broader Africa
rollout

Future SIP service: AI-assisted private tutoring with video/audio calls similar to Meet, Teams and Zoom, plus a shared maths/text canvas for tutors and learners.

Founder advantage

Technical founder with QA, AI/data, engineering and teaching experience.

SharperTime

Google Africa Applied AI Lab

Lesego Mochaule

**Founder & Director
Sharper X (Pty) Ltd**

Johannesburg, South Africa

BSc (Hons) Big Data Analytics
BSc Geo-Informatics
National Diploma Electrical Engineering
ISTQB Certified Tester Foundation Level

- **Software QA**

Functional, regression, usability, web and mobile testing; Appium and Selenium experience.

- **AI & data capability**

Python, C#, Java, SQL, TensorFlow, PyTorch and data-analysis tools.

- **Education link**

Former computer-science lab assistant and practical learner-support experience.

- **Builder mindset**

Can build, test, iterate and launch the product rather than only outsource development.

The founder profile matches SharperTime's immediate needs: build, test, launch, measure and improve.



Personalised AI tutoring that can scale from smartphones to schools across Africa.

Why Google Africa Applied AI Lab

Technical support, venture-building and Google AI tooling can help SharperTime improve multimodal performance, scale lesson content, refine questions and expand distribution.

Sharper Solutions. Sharper Futures.

Sharper X (Pty) Ltd

